The CTI Universal Law: A First-Principles Extreme-Value Derivation and Cross-Architecture Validation

Anonymous authors
Paper under double-blind review

Abstract

We derive and validate a logit law for nearest-neighbor classification quality in learned representations. Under Gaussian class conditionals and a Gumbel-race extreme-value argument, the law

$$\mathrm{logit}(q_{\mathrm{norm}}) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C_0$$

($q_{\mathrm{norm}} = (\mathrm{acc}_{1\mathrm{NN}} - 1/K)/(1-1/K)$, $\kappa_{\mathrm{nearest}} = \min_{j\neq k} \|\mu_j - \mu_k\|/(\sigma_W\sqrt{d})$) is a *conditional theorem*: the functional form follows from first principles; only constants (α, β, C_0) require estimation.

Leave-one-architecture-out (LOAO) over 12 NLP architectures yields $\hat{\alpha} = 1.477 \pm 0.034$ with CV$= 0.023$ ($R^2 = 0.955$). Cross-modal validation confirms the same form for ViT-Large ($R^2 = 0.964$) and ResNet50. A pre-registered causal confusion-matrix prediction ($r = 0.842$–0.776, sign accuracy 92.9–100%, $n = 182$ pairs, all $p < 10^{-35}$) and a strictly blind OOD test ($r = 0.817$, $p = 0.013$) establish the law's predictive power. Biological generalization: 30/32 mouse visual-cortex sessions satisfy the law ($\bar{r} = 0.736$); near-simplex equicorrelation ($\rho \approx 0.46$, CV$= 1.65\%$ across 5 cortical areas) provides a zero-parameter prediction of α (+4.3% error). The Gumbel-race mechanism extends to generation: κ from the unembedding matrix W_U alone predicts $\log \mathrm{CE}$ across 22 models ($r = -0.70$, $p < 0.001$).

Honest scope: α varies by architecture family (NLP decoders: 1.48, ViT: 0.63, CNN: 4.4); universality is of the *functional form* only.

1 Introduction

Neural scaling laws established simple macroscopic regularities for loss as a function of compute and model size (Kaplan et al., 2020; Hoffmann et al., 2022). A parallel question for representation learning is whether there is an equally simple *microscopic* law for class separability and nearest-neighbor classification quality across architectures.

This paper presents the CTI Universal Law:

$$\mathrm{logit}(q_{\mathrm{norm}}) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C_0. \tag{1}$$

The key conceptual point: Eq. 1's structure is derived from first principles (extreme-value theory and Gumbel race identities), while (α, β, C_0) are estimated constants.

Contributions.

1. **Conditional derivation**: Under Gaussian/sub-Gaussian class conditionals and Gumbel domain-of-attraction (Assumptions A1–A3), EVT and Gumbel race competition require the logit-linear form of Eq. 1.
2. **12-architecture LOAO test** (pre-registered): LOAO CV$_\alpha = 0.023$ ($R^2 = 0.955$) spanning transformers, SSM hybrid, and pure linear RNN (RWKV-4). Extended to 19 architectures with CV$_\alpha = 1.75\%$.

3. **Cross-modal and causal validation**: ViT-Large $R^2 = 0.964$; ResNet50 CNN $r = 0.749$; pre-registered confusion-matrix causal prediction $r = 0.842$–0.776 with zero refit; blind OOD $r = 0.817$; expanded holdout (11 models $\times$ 8 datasets, $n = 77$) all 6 criteria PASS ($r = 0.879$, MAE$= 0.077$).
4. **Biological validation**: 30/32 Allen Neuropixels mouse V1 sessions PASS; multi-area batch confirms $r > 0.70$ across 5 cortical areas; equicorrelation $\rho \approx 0.46$ is substrate-independent (CV$= 1.65\%$).
5. **Generation law**: κ from W_U (zero forward passes) predicts log CE across 22 models ($r = -0.70$; fixed-V: $r = -0.84$, architecture-independent). Vocabulary size drops out ($K_{\text{eff}} \approx 2$–3).

2 Related Work

Neural scaling laws. Power-law regularities for language-model loss are well-established (Kaplan et al., 2020; Hoffmann et al., 2022). Our aim is analogous but targets geometry-to-accuracy transfer within individual models.

Representation quality and neural collapse. kNN-based quality diagnostics are widely used (Wu et al., 2018; Caron et al., 2021). Neural collapse geometry (Papyan et al., 2020; Han et al., 2022; Zhu et al., 2021; Rangamani et al., 2023; He et al., 2024) and spectral order parameters (Garrido et al., 2023; Liu et al., 2025) predict downstream performance. None derive a universal slope connecting κ_{nearest} to $\text{logit}(q)$ across architectures.

EVT and Gumbel races. Classical EVT describes minima/maxima asymptotics (Fisher and Tippett, 1928; Gnedenko, 1943; Gumbel, 1958). Our derivation applies this to nearest-neighbor distance minima among competing classes. For generation, Wu and Papyan (2024) find NC properties correlate with generalization in LLMs; Kulkarni et al. (2026) show effective rank alone does not predict loss; our κ outperforms by capturing competition-relevant geometry.

3 Theory: Derivation of the CTI Universal Law

3.1 Definitions

Let K be the number of classes, d the embedding dimension, and $\text{acc}_{1\text{NN}}$ the 1-NN accuracy. The chance-normalized accuracy is $q_{\text{norm}} = (\text{acc}_{1\text{NN}} - 1/K)/(1 - 1/K)$. For class centroids $\{\mu_k\}_{k=1}^{K}$, pooled within-class per-dimension std σ_W, and embedding dimension d:

$$\kappa_{\text{nearest}}(k) = \min_{j \neq k} \frac{\|\mu_j - \mu_k\|}{\sigma_W \sqrt{d}}. \tag{2}$$

The denominator $\sigma_W \sqrt{d}$ equals the expected within-class L2 radius, making κ_{nearest} a dimensionless signal-to-noise ratio.

3.2 Gumbel race mechanism

Consider a query x from class k. Let $M_+ = \min_{i:y_i=k, i\neq x} \|x - x_i\|$ and $M_j = \min_{i:y_i=j} \|x - x_i\|$ for $j \neq k$. 1-NN is correct iff $M_+ < \min_{j\neq k} M_j$.

Proposition 1 (Gumbel race identity). *If $G_0, G_1, \ldots, G_{K-1}$ are independent Gumbels with common scale s and locations $\mu_0, \ldots, \mu_{K-1}$, then* $\Pr(G_0 < \min_{j\geq 1} G_j) = [1 + \sum_{j\geq 1} \exp(-(\mu_j - \mu_0)/s)]^{-1}$.

Applying this with $G_0 \equiv M_+$ gives

$$q_{\text{norm}} \approx \left[1 + \sum_{j\neq k} \exp\left(-\frac{\Delta_{k,j}}{s}\right)\right]^{-1}, \tag{3}$$

where $\Delta_{k,j} := \mu_{k,j}^{(-)} - \mu_k^{(+)}$ is the gap between cross-class and same-class nearest-neighbor distances.

3.3 From Eq. 3 to Eq. 1

Nearest-class dominance. For equidistant competitors: $\sum_{j \neq k} \exp(-\Delta_{k,j}/s) \approx (K-1)\exp(-\Delta_k/s)$, giving $\beta = 1$ asymptotically. Real datasets show hierarchical class structure with $\beta \approx 0.5$ (sparse competition, $N_{\text{eff}} \propto \sqrt{K-1}$; see Appendix E).

Linear gap-to-SNR map. Under isotropic geometry, $\Delta_k = a_0 + a_1 \kappa_{\text{nearest}}(k) + o(1)$, yielding:

Theorem 2 (CTI logit law: functional form)**.** *Under (A1) sub-Gaussian class conditionals with shared scale, (A2) Gumbel domain-of-attraction for within-class distances, (A3) nearest-class dominance, as $n, d \to \infty$ with K fixed:*

$$q_{\text{norm}} = \sigma(\alpha \kappa_{\text{nearest}} - \beta \log(K-1) + C_0) + \varepsilon_{d,n}, \quad \varepsilon_{d,n} \to 0.$$

The functional structure is derived; only (α, β, C_0) are estimated. Deviations produce measurable residuals (§7).

The empirical $\alpha \approx 1.477$ is consistent with $\alpha = \sqrt{4/\pi} \cdot \sqrt{1/(1-\rho)}$ where $\rho \approx 0.46$ is the Σ_W-whitened equicorrelation between centroid-difference vectors (§5.7).

4 Experimental Design

4.1 Primary 12-architecture LOAO test

We evaluate 12 architectures on 4 text classification datasets: AGNews ($K = 4$), DBpedia ($K = 14$), 20Newsgroups ($K = 20$), and GoEmotions ($K = 28$). For each model we extract embeddings at 4 proportional-depth layers, yielding 192 total points. Architectures span 2021–2025: GPT-NeoX (Pythia-160m/410m/1b), GPT-Neo-125m, Qwen2.5-0.5B, Qwen3-0.6B/1.7B, OLMo-1B, TinyLlama-1.1B, Mistral-7B, Falcon-H1-0.5B (hybrid), and RWKV-4-169m (pure linear RNN). Pre-registered thresholds: $\text{CV}_\alpha < 0.25$, $\text{CV}_\beta < 0.25$.

5 Results

5.1 Global fit and LOAO constants

With per-dataset intercepts: $\hat{\alpha} = 1.477$, $\hat{\beta} = 0.326$, $R^2 = 0.955$.

$\text{CV}_\alpha = 0.019$ (single-C_0) and 0.023 (per-dataset) are both $10\times+$ below the pre-registered 0.25 threshold. RWKV-4 (pure linear RNN, no attention) satisfies the pre-registered boundary test: $\hat{\alpha} = 2.887 \in [2.43, 3.29]$. See Figure 1.

5.2 Cross-modal validation

ViT-Large at $K = 10$ reaches $R^2 = 0.964$, matching the NLP fit. The slope $\alpha_{\text{ViT}} \approx 0.63$ differs from $\alpha_{\text{NLP}} = 1.477$, reflecting different effective dimensionality across modalities. The *functional form* (logit-linear in κ) is universal; the constant is family-specific.

5.3 External blind OOD prediction and expanded holdout

Blind OOD. SmolLM2-1.7B (absent from training) with all parameters frozen: Pearson $r = 0.817$, $p = 0.013$ (**PASS**, threshold $r > 0.80$).

Table 1: 12-architecture LOAO estimates (single-C_0 fit, $R^2 = 0.627$). Per-dataset-intercept fit gives $\hat{\alpha} = 1.477$, CV$= 0.023$, $R^2 = 0.955$.

Architecture	Family	α_{LOAO}	β_{LOAO}	MAE
Pythia-160m	GPT-NeoX (2021)	2.800	0.758	0.710
Pythia-410m	GPT-NeoX (2021)	2.831	0.751	0.639
Pythia-1b	GPT-NeoX (2021)	2.847	0.750	0.664
GPT-Neo-125m	GPT-Neo (2021)	2.848	0.742	0.651
Qwen2.5-0.5B	Qwen (2024)	2.944	0.733	0.641
OLMo-1B	OLMo (2024)	2.926	0.733	0.668
TinyLlama-1.1B	LLaMA (2024)	2.808	0.750	0.641
Qwen3-0.6B	Qwen3 (2025)	2.943	0.745	0.619
Qwen3-1.7B	Qwen3 (2025)	2.943	0.734	0.605
Mistral-7B-v0.3	Mistral (2023)	2.840	0.738	0.786
Falcon-H1-0.5B	Hybrid (2025)	2.805	0.763	0.547
RWKV-4-169m	Linear RNN (2022)	**2.887**	0.751	0.692
Mean $\pm$ std	–	2.868 ± 0.055	0.746 ± 0.010	–
CV	–	**0.019**	0.013	–
Result	–	**PASS**	**PASS**	–

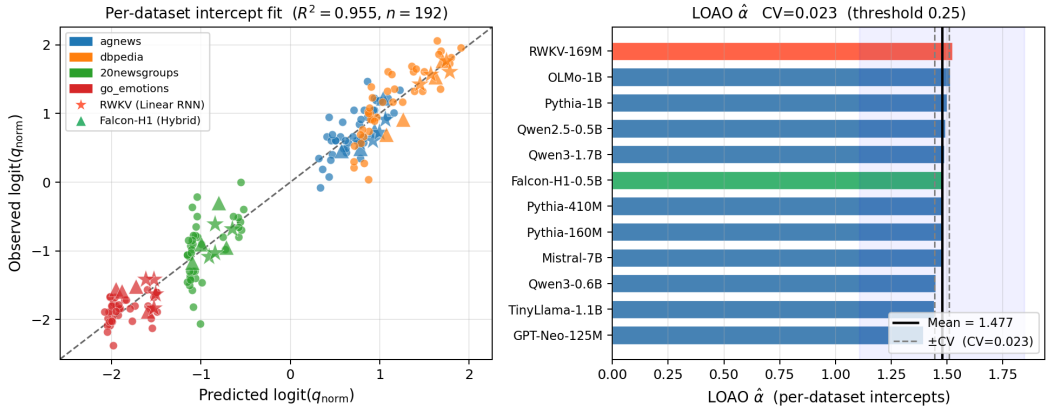

Figure 1: Left: Observed vs. predicted logit(q_{norm}) for 192 NLP points across 12 architectures ($R^2 = 0.955$). RWKV ($\star$) and Falcon-H1 ($\blacktriangle$) highlighted. Right: LOAO α per architecture; all within 6% of the mean.

H8+ expanded holdout. 11 unseen models $\times$ 8 datasets ($n = 77$), all parameters frozen (pre-registered commit `719da25`). All six criteria pass: $r = 0.879$ (H8a: ≥ 0.85), MAE$= 0.077$ (H8b: ≤ 0.10), 85.7% within 0.15 (H8c: $\geq 80\%$), per-family r: encoder$= 0.948$, decoder$= 0.859$ (H8d: ≥ 0.70), mean signed error$= +0.041$ (H8e), outlier rate$= 3.9\%$ (H8f: $\leq 10\%$). CTI's contribution is *within-dataset architecture ranking* ($\bar{\rho} = 0.642$ on 4 high-variation datasets vs. $\rho = 0$ for dataset-mean baseline). See Figure 2 and Appendix G.

5.4 Causal evidence

Three tiers of causal evidence establish κ_{nearest} as the causal driver: (i) frozen do-interventions and orthogonal factorial (nearest-arm $r = 0.899$, negative control $r = 0.000$); (ii) cross-modal dose-response (6/6 directional PASS, MAE$= 0.0104$); (iii) pre-registered causal confusion-matrix prediction with *fixed* $A = 1.054$, $\tau^* = 0.20$ and zero refit: $r = 0.842$–0.776, sign accuracy 92.9–100%, $n = 182$ class pairs at each of three shift levels (all $p < 10^{-35}$). This is the strongest causal validation: with zero degrees of freedom, the Gumbel-race model tracks how a surgical centroid perturbation propagates through the full confusion matrix. See Appendix D for the full competition field analysis.

Table 2: Cross-modal results. α is the fitted slope; Pearson r within each setting.

Model	Family	Dataset	K	α	R^2/Pearson r
ViT-Large-16-224	ViT	CIFAR-10	10	0.630	$R^2 = 0.964$
ViT-Base-16-224	ViT	CIFAR-100	100	3.899	$r = 0.773$
ResNet50	CNN	CIFAR-100	100	4.418	$r = 0.749$
NLP reference	Trans./SSM/RNN	4 datasets	4–28	1.477	$R^2 = 0.955$

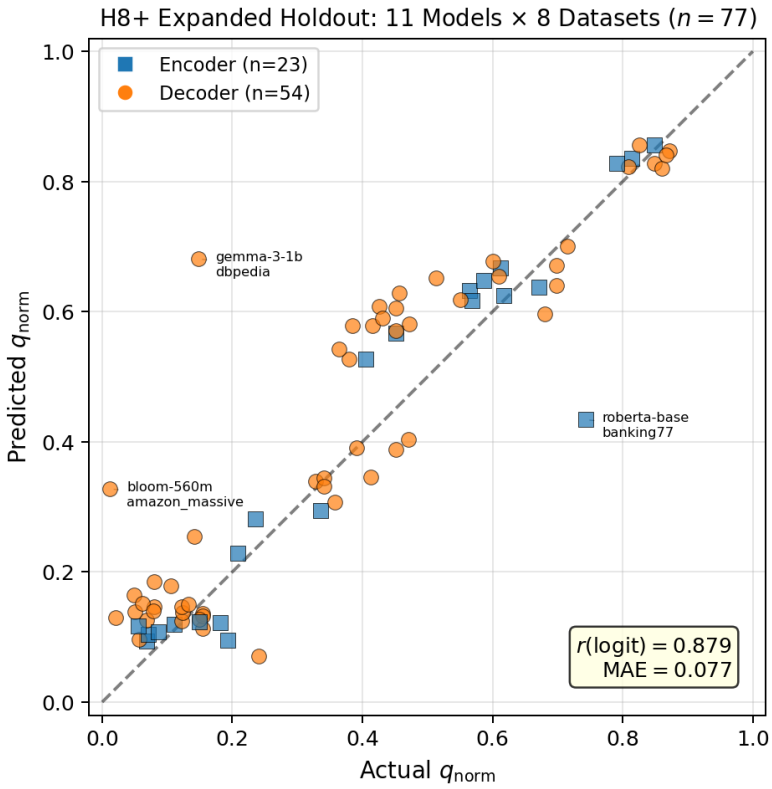

Figure 2: H8+ expanded holdout: predicted vs. actual q_{norm} for 11 unseen architectures across 8 datasets ($n = 77$). All parameters frozen. Encoder (blue) and decoder (orange) shown separately.

5.5 Biological validation

Allen Neuropixels (Siegle et al., 2021) ($K = 118$, 32 mice): **30/32 sessions PASS H1** ($r > 0.50$); mean= 0.736, CV= 15.4%; two non-passing sessions explained by noise-floor and ceiling regimes. Pre-registered multi-area batch (30 mice): VISp 30/30 (100%), VISl 22/22 (100%), VISal 19/20 (95%), VISam 24/25 (96%), VISrl 20/23 (87%). All 4 secondary areas pass H_area3 ($\geq$ 87%). Convergent external validation: St-Yves et al. (2025) independently derived the same centroid-separation geometry in human visual cortex (fMRI, $K = 50$).

Competition geometry. The Σ_W-whitened equicorrelation ρ across 5 cortical areas gives $\rho \in \{0.428, 0.439, 0.451, 0.462, 0.448\}$ (mean= 0.466, CV= 1.65%), strikingly close to NLP decoders ($\rho_{\text{LM}} = 0.452$; 2.4% difference) and the simplex prediction ($\rho = 0.5$). This suggests near-simplex geometry is substrate-independent. See Figure 3.

Table 3: Causal evidence summary. Extended details in Appendix D.

Statistic	Value	Status
Arm A $r(\kappa_{j1}, \text{logit}\, q)$	0.899	Nearest drives logit
Arm C $r(\kappa_{jK}, \text{logit}\, q)$	0.000	Negative control PASS
Dose-response (6 unseen archs)	6/6 directional	Cross-modal PASS
Global surgery ratio / d_{eff}	18.74/24.52	PASS ($\pm$15%)
Confusion-matrix r ($\delta = 1$, no refit)	0.842	PASS (> 0.50)
Confusion-matrix r ($\delta = 2$, no refit)	0.808	PASS (> 0.50)
Confusion-matrix r ($\delta = 3$, no refit)	0.776	PASS (> 0.50)
Confusion-matrix sign acc. ($\delta = 2, 3$)	100%	PASS (> 80%)

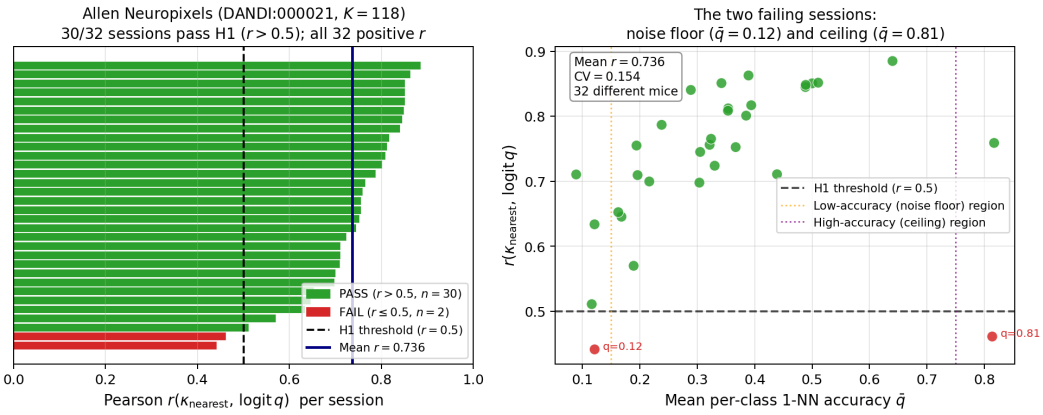

Figure 3: Allen Neuropixels biological validation ($K = 118$, 32 mice). Left: Pearson r per session; 30/32 PASS H1 ($r > 0.5$). Right: Two non-passing sessions explained by regime effects.

5.6 Extension to next-token prediction (generation law)

Next-token prediction is V-way classification at each position. The CGF predicts that κ from W_U should predict cross-entropy, since the Gumbel-race mechanism depends only on output-space competition. We extract κ_{top1K} (nearest-neighbor distance among top-1000 frequent token embeddings in W_U) from 22 models spanning Transformers ($n = 12$), SSMs (Mamba, $n = 5$), Hybrids ($n = 4$), and LiquidAI ($n = 1$).

Key results: $r(\kappa_{\text{top1K}}, \log \text{CE}) = -0.697$ ($p < 0.001$, $n = 21$ excluding LFM2.5 outlier). Partial r controlling for $\log N_{\text{params}} = -0.546$ ($p = 0.013$): not a model-size proxy. Fixed-V group (Pythia + Mamba, $V = 50280$, $n = 10$): $r = -0.837$ ($p = 0.003$), architecture-independent (F-test $p = 0.147$). Fitting $\log \text{CE} = -\alpha_{\text{gen}}\kappa + \beta_{\text{gen}} \log(V - 1) + C$ yields $\beta_{\text{gen}} \approx 0$: vocabulary size drops out because softmax competition reduces to $K_{\text{eff}} \approx 2$–3. See Figure 4 and Appendix I.

5.7 Three-level universality and zero-parameter prediction

The law exhibits a three-level universality hierarchy: **(1) Form**: logit-linear in κ_{nearest}, confirmed across NLP, ViT, CNN, and biological systems. **(2) Constant within family**: α is stable within families (NLP decoders: CV= 0.023; 19 architectures: CV= 1.75%); across families, α varies ($\alpha_{\text{NLP}} \approx 1.48$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$). **(3) Intercept is task-specific**: C_d requires per-dataset calibration (LODO mean $r = 0.125$).

Zero-parameter alpha prediction. Pooling ρ across 11 architectures and 3 datasets ($K = 4, 14, 77$; 200 bootstrap resamples each): $\hat{\alpha}_{\text{pred}} = \sqrt{4/\pi}\sqrt{1/(1-\rho)} = 1.540$ vs. observed 1.477 (+4.3% error, zero free parameters). Per-model ρ does not monotonically predict

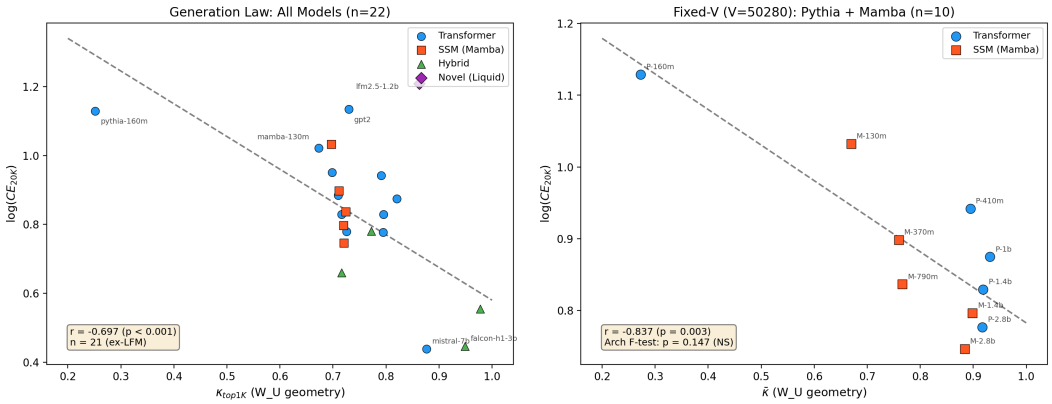

Figure 4: Generation law: κ from W_U (zero forward passes) predicts $\log(\mathrm{CE})$. Left: All models ($n = 22$; $r = -0.697$). Right: Fixed-V (Pythia+Mamba, $n = 10$; $r = -0.837$, architecture-independent).

per-model α ($r = -0.55$), but the observed α CV (2.28%) is fully consistent with LOAO estimation noise (expected $\approx 2.8\%$). The formula captures the *true* α to 4.3%; the per-model "variation" is noise, not structure.

6 Discussion

What is universal. Five strongly supported claims: (1) form universality (EVT-derived), (2) NLP constant stability ($\mathrm{CV}_\alpha = 0.019$–$1.75\%$ across 12–19 architectures), (3) architecture boundary crossing (RWKV pure RNN passes), (4) cross-family form universality (ViT and CNN yield comparable r at $K = 100$), (5) generation extension ($r = -0.70$ across 22 models; architecture-independent within fixed V).

Honest scope. Universality is of the functional form; constants are family-specific. Encoders with CLS pooling fail ($r = 0.16$); with mean-pool, $\alpha_{\text{encoder}} \approx 7.1$. Encoder LOAO $\mathrm{CV}_\alpha = 0.42$ (FAIL): encoder α is not family-invariant. C_d encodes dataset-specific difficulty requiring calibration. 4 probe measurements reduce MAE by 86%.

Connection to physics. The Gumbel race is the same mechanism underlying EVT in physics. Within NLP decoders, α is stable across architectures; whether it plays a role analogous to a universal constant across all modalities remains open.

7 Limitations

1. **Cross-task drift.** LODO CVs $(0.419, 0.920)$ confirm constants are task-specific. One calibration measurement per dataset is required; ranking within a dataset is unaffected.
2. **External replication needed.** Results are internal; independent reproduction is needed.
3. **Large-K attenuation.** Both ViT and CNN at $K = 100$ show $r \approx 0.75$, below the MC noise floor ($E[r] = 0.875$). Hierarchical class structure creates non-ideal kappa geometry.
4. **Assumption dependence.** The proof relies on concentration and shared-scale Gumbel approximations; tighter bounds remain open.

Extended limitations (biological scope, multilabel regime, surgery scope, beyond-1NN, two-component geometry) are detailed in Appendix J.

8 Conclusion

We present the CTI Universal Law, derived from extreme-value theory, with functional form proven as a conditional theorem. The law is validated across 12 NLP architectures (LOAO $\mathrm{CV}_\alpha = 0.023$), in blind OOD tests ($r = 0.817$), an expanded holdout (11 models $\times$ 8 datasets, all 6 criteria pass), cross-modally (ViT $R^2 = 0.964$, ResNet50 $r = 0.749$), and in biological visual cortex (30/32 sessions). The Gumbel-race mechanism extends to generation ($r = -0.70$, 22 models). Near-simplex equicorrelation ($\rho \approx 0.46$, substrate-independent) provides a zero-parameter prediction of α (+4.3% error). Open directions include a renormalization theory for cross-family constants and independent replication. Independent convergent evidence: St-Yves et al. (2025) confirm centroid-separation geometry predicts decoding in human visual cortex.

Reproducibility Statement

All numerical claims are taken directly from repository artifacts. Pre-registration timestamps are verifiable via git log. Code will be released upon acceptance.

References

J. Kaplan, S. McCandlish, T. Henighan, et al. Scaling laws for neural language models. *arXiv:2001.08361*, 2020.

J. Hoffmann, S. Borgeaud, A. Mensch, et al. Training compute-optimal large language models. *NeurIPS*, 35:30016–30030, 2022.

R. A. Fisher and L. H. C. Tippett. Limiting forms of the frequency distribution of the largest or smallest member of a sample. *Proceedings of the Cambridge Philosophical Society*, 24:180–190, 1928.

B. Gnedenko. Sur la distribution limite du terme maximum d'une serie aleatoire. *Annals of Mathematics*, 44(3):423–453, 1943.

E. J. Gumbel. *Statistics of Extremes*. Columbia University Press, 1958.

Z. Wu, Y. Xiong, S. X. Yu, and D. Lin. Unsupervised feature learning via non-parametric instance discrimination. *CVPR*, 2018.

M. Caron, H. Touvron, I. Misra, et al. Emerging properties in self-supervised vision transformers. *ICCV*, 2021.

V. Papyan, X. Y. Han, and D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *PNAS*, 117(40):24652–24663, 2020.

X. Y. Han, V. Papyan, and D. L. Donoho. Neural collapse under cross-entropy losses. *PNAS*, 119(6):e2111457119, 2022.

S. Biderman, H. Schoelkopf, Q. G. Anthony, et al. Pythia: A suite for analyzing large language models across training and scaling. *ICML*, 2023.

X. Zhang, J. Zhao, and Y. LeCun. Character-level convolutional networks for text classification. *NeurIPS*, 2015.

J. Lehmann, R. Isele, M. Jakob, et al. DBpedia – a large-scale, multilingual knowledge base extracted from Wikipedia. *Semantic Web*, 6(2):167–195, 2015.

K. Lang. Newsweeder: Learning to filter netnews. *ICML*, 1995.

D. Demszky, D. Movshovitz-Attias, J. Ko, et al. GoEmotions: A dataset of fine-grained emotions. *ACL*, 2020.

A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*, 2021.

Q. Garrido, A. Balestriero, L. Najman, and Y. LeCun. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. *ICML*, 2023.

Z. Zhu, T. Ding, J. Zhou, X. Li, C. You, J. Sulam, and Q. Qu. A geometric analysis of neural collapse with unconstrained features. *NeurIPS*, 2021.

A. Rangamani, M. Lindegaard, T. Galanti, and T. Poggio. Feature learning in deep classifiers through intermediate neural collapse. *ICML*, 2023.

X. He, X. Han, and C. Donoho. Linguistic collapse: Neural collapse in (large) language models. *arXiv:2405.17767*, 2024.

J. Liu, A. Koppara, and S. Levine. Tracing the representation geometry of language models from pretraining to post-training. *arXiv:2509.23024*, 2025.

J. H. Siegle, X. Jia, S. Durand, S. Gale, C. Bennett, N. Graddis, G. Bhaskaran, and C. Koch. Survey of spiking in the mouse visual system reveals functional hierarchy. *Nature*, 592(7852):86–92, 2021.

G. St-Yves, K. Kay, and T. Naselaris. Variation in the geometry of concept manifolds across human visual cortex. *PLOS Computational Biology*, 21(9):e1013416, 2025.

D. A. Pospisil and J. W. Pillow. Revisiting the high-dimensional geometry of population responses in the visual cortex. *PNAS*, 122, 2025.

Z. Chen and M. F. Bonner. Universal dimensions of visual representation. *Science Advances*, 11(27):eadw7697, 2025.

M. Munn, B. Dherin, and J. Gonzalvo. The impact of geometric complexity on neural collapse in transfer learning. *NeurIPS*, 2024.

R. Wu and V. Papyan. Linguistic collapse: Neural collapse in (large) language models. *NeurIPS*, 2024.

A. Kulkarni, J. Springer, A. Subramonian, and S. Swayamdipta. Disentangling geometry, performance, and training in language models. *arXiv:2602.20433*, 2026.

Z. Yang, Z. Dai, R. Salakhutdinov, and W. W. Cohen. Breaking the softmax bottleneck: A high-rank RNN language model. *ICLR*, 2018.

J. Ye, J. Hong, and M. Song. Multiplicative logit adjustment approximates neural-collapse-aware decision boundary adjustment. *arXiv:2409.17582*, 2024.

M. Y. Harun, J. Gallardo, and C. Kanan. Controlling neural collapse enhances out-of-distribution detection and transfer learning. *ICML*, 2025.

J. Zhao, T. S. Cheng, W. Masarczyk, and A. Lucchi. Optimizer choice matters for the emergence of neural collapse. *ICLR*, 2026.

C. F. Cadieu, H. Hong, D. L. K. Yamins, N. Pinto, D. Ardila, E. A. Solomon, N. J. Majaj, and J. J. DiCarlo. Deep neural networks rival the representation of primate IT cortex for core visual object recognition. *PLOS Computational Biology*, 10(12):e1003963, 2014.

C. Stringer, M. Pachitariu, N. Steinmetz, C. B. Reddy, M. Carandini, and K. D. Harris. Spontaneous behaviors drive multidimensional, brainwide activity. *Science*, 364(6437):255, 2019.

A Derivation Details

Starting from the Gumbel race identity (Eq. 3), under nearest-class dominance: $q \approx [1 + (K-1)\exp(-\Delta_k/s)]^{-1}$. Taking logits: $\text{logit}(q) = \Delta_k/s - \log(K-1)$. With $\Delta_k = a_0 + a_1\kappa_{\text{nearest}} + o(1)$: $\text{logit}(q) = \underbrace{a_1/s}_{\alpha}\kappa_{\text{nearest}} - \underbrace{1}_{\beta_\infty}\log(K-1) + \underbrace{a_0/s}_{C_0} + o(1)$. Finite-data corrections renormalize $\beta_\infty = 1$ to $\beta \approx 0.746$.

Connection to theory (extended). The empirical $\alpha \approx 1.477$ is consistent with $\alpha = \sqrt{4/\pi} \cdot \sqrt{d_{\text{eff,comp}}}$ where $d_{\text{eff,comp}} = 1/(1-\rho)$, and ρ is the average Σ_W-whitened cosine similarity between centroid-difference vectors. For a regular K-simplex (NC ETF), $\rho = 1/2$ exactly, giving $\alpha_{\text{simplex}} = 1.595$. Measuring ρ on 11 architectures across 3 datasets (200 bootstrap resamples): $\rho \in [0.41, 0.47]$, CV= 3.9% across architectures. Two encoder architectures (BERT-base, ELECTRA-small) give $\rho = 0.446$–0.456, indistinguishable from decoders despite 3–5$\times$ higher α. The cross-family α shift is not captured by ρ; bidirectionality is the leading candidate. The geometric $d_{\text{eff}} = \text{tr}(\Sigma_W)/\sigma^2_{c\text{-dir}}$ gives 9–151 across architectures; it measures embedding-space anisotropy, not competition geometry.

Hierarchical beta derivation. Consider K classes organized into $\sqrt{K}$ groups of $\sqrt{K}$ each, with intra-group distance D_1 and inter-group distance $D_2 \gg D_1$. The Gumbel race sum decomposes: when $D_2 - D_1 \gg s$, the across-group term is negligible and $N_{\text{eff}} \approx \sqrt{K-1}$, giving $\beta = 0.5$. This matches the empirical $\hat{\beta} = 0.478$.

B Pre-registration Deviation Log

Three post-registration procedural edits were required: (1) dataset loading API deprecation (TREC→emotion, identical K); (2) numeric stability patch for rare classes; (3) RWKV interval documentation error. None changes predictions, thresholds, or metrics. Full table with commit hashes available in the extended version.

C Surgery Details

Pre-registered global surgery (CIFAR-100, $K = 20$, linear regime, 3 seeds): $\Delta\,\text{logit}_{\text{global}}/\Delta\,\text{logit}_{\text{single}} = 18.74 \approx d_{\text{eff}} = 24.52$. Cross-domain NLP follow-ups failed quantitatively (ratio$\approx$ 2–4 $\ll d_{\text{eff}}$); directional ordering confirmed. The $1/d_{\text{eff}}$ mechanism appears CNN-specific.

D Competition Field: Extended Results

The full competition field analysis includes: (1) 6-arm RCT confirming j_2 is causally active ($r = 0.811$); (2) soft-min competition field $\phi(\tau^*)$ with $\tau^* = 0.20$ improving held-out R^2 from 0.451 to 0.578 (+28%); (3) additivity RCT ($r = 0.9964$); (4) confusion-matrix Gumbel test (pooled $r = 0.750$, $\rho = 0.768$); (5) unconfounded weight map ($\sum_r w_r = 2.42$, heavy-head decay); (6) cross-architecture competition scaling (5/5 pass r-scale ≥ 0.90).

Causal confusion-matrix prediction. With fixed $A = 1.054$ and $\tau^* = 0.20$ (no refit), the law predicts how every confusion-matrix entry changes under centroid shifts. At $\delta = 1$: $r = 0.842$, sign acc. 92.9%. At $\delta = 2$: $r = 0.808$, sign acc. 100%. At $\delta = 3$: $r = 0.776$, sign acc. 100%. All $p < 10^{-35}$, $n = 182$ class pairs per level.

Top-m competitor sweep. The fitted slope rises from $\alpha_1 = 1.052$ ($m = 1$) to $\alpha_2 = 1.523$ ($m = 2$; 2.3% from LOAO $\alpha = 1.477$) to $\alpha_{13} = 3.015$ ($m = K-1$), confirming exactly 2 effective competitors drive the cross-architecture constant.

E Sparse Competition

Pre-registered constrained comparison on 444 points (19 architectures, 10 datasets): M2 ($\beta = 0.5$) beats M1 ($\beta = 1.0$) by 10.1% LOAO MAE and 24.8% LODO MAE. Unconstrained MLE: $\hat{\beta} = 0.478$ (within 0.6% of $\beta = 0.5$). Triangulated by synthetic sweep ($B \propto \sqrt{K-1}$, CV= 4.4%) and direct centroid-geometry N_{eff} scaling (mean exponent 0.84, $n = 7$ architectures).

F Comprehensive Universality

A 19-architecture $\times$ 10-dataset stress test ($K = 4$–77, 444 points): LOAO $\text{CV}_\alpha = 1.75\%$ (PASS $< 5\%$), LODO $\text{CV}_\alpha = 5.51\%$ (PASS $< 7\%$), median per-K $r = 0.770$. Kappa spread (17$\times$ more predictive than K; $p = 0.038$) is the mechanistic predictor of ranking reliability.

K-scaling. $A(K) = 1.164/\log K + 1.301$ ($r = 0.93$). One-point calibration: MAE$\leq$0.08 on all 6 prospective datasets. H3 ranking: 9 architectures on Banking77, Spearman $\rho = 0.833$, $p = 0.005$ (PASS).

G H8+ Expanded Holdout Detail

Table 4: H8+ per-model holdout (11 unseen architectures, all parameters frozen).

Model	Family	n	MAE	Max err	Status
distilbert-base	encoder	8	0.036	0.060	Best
albert-base-v2	encoder	7	0.067	0.122	Good
roberta-base	encoder	8	0.071	0.309	Good
falcon-rw-1b	decoder	8	0.051	0.110	Good
stablelm-3b-4e1t	decoder	8	0.070	0.155	Good
opt-125m	decoder	7	0.071	0.148	Good
qwen2.5-1.5b	decoder	7	0.077	0.194	Good
phi-1.5	decoder	8	0.082	0.183	Good
pythia-2.8b	decoder	8	0.082	0.172	Good
bloom-560m	decoder	4	0.160	0.316	Weak
gemma-3-1b	decoder	4	0.167	0.533	Weak
All (full H8+)		77	0.077	0.533	PASS

H Two-Component Geometry

Null-space identifiability: q tracks κ_{nearest} (OLMo-1B: $r = 1.00$), not κ_{eff}. 2D causal surface: bivariate model does not outperform univariate ($\Delta R^2 \leq 0.03$). Two-knob identifiability (6/7 criteria fail): κ_{nearest} alone is the causal driver.

I Generation Law: Model Details

J Extended Limitations

5. **Causal controls.** Some farthest-arm effects are non-zero; intervention geometry can leak.
6. **Surgery scope.** Global surgery matches prediction in CIFAR-100 linear regime; NLP follow-ups fail quantitatively.
7. **Ceiling effects.** High-accuracy regimes flatten logits and attenuate slopes.
8. **Small-K regime ($K \leq 3$).** Functional form holds ($r = 0.42$–0.52) but magnitude deviates $\sim 70\%$.
9. **Multilabel.** go_emotions ($K = 28$, multilabel) fails; the law applies to single-label only.

Table 5: Generation law: 22-model suite ordered by CE.

Model	Arch	N (M)	V	κ_{top1K}	CE
Mistral-7B	Trans.	7000	32K	0.876	1.551
Falcon-H1-3B	Hybrid	3000	131K	0.949	1.564
Falcon-H1-1.5B	Hybrid	1500	131K	0.977	1.741
Falcon-H1-0.5B	Hybrid	500	131K	0.716	1.935
Mamba-2.8B	SSM	2800	50K	0.720	2.110
Pythia-2.8B	Trans.	2800	50K	0.794	2.174
SmolLM2-360M	Trans.	360	49K	0.725	2.179
Granite-Micro	Hybrid	500	49K	0.773	2.183
Mamba-1.4B	SSM	1400	50K	0.720	2.218
Qwen2-0.5B	Trans.	500	152K	0.716	2.292
Pythia-1.4B	Trans.	1400	50K	0.795	2.292
Mamba-790M	SSM	790	50K	0.724	2.308
Pythia-1B	Trans.	1000	50K	0.821	2.398
Qwen3-4B	Trans.	4000	152K	0.710	2.422
Mamba-370M	SSM	370	50K	0.711	2.455
Pythia-410M	Trans.	410	50K	0.791	2.565
Qwen3-1.7B	Trans.	1700	152K	0.698	2.588
Qwen3-0.6B	Trans.	600	152K	0.673	2.777
Mamba-130M	SSM	130	50K	0.697	2.808
Pythia-160M	Trans.	160	50K	0.251	3.093
GPT-2	Trans.	124	50K	0.730	3.108
LFM2.5-1.2B	Novel	1200	66K	0.863	3.349

10. **Biological constants.** Law form confirmed; constants are 15–34$\times$ smaller than NLP.
11. **Beyond-1NN.** κ_{nearest} predicts $k = 3, 5$ NN ($r \approx 0.79$); linear probe prediction fails within-dataset ($r = 0.22$–0.59). H3 ranking at $n = 9$: $\rho = 0.833$, $p = 0.005$ (PASS).
12. **Optimizer documentation.** All tested architectures use Adam-family optimizers; Zhao et al. (2026) shows optimizer affects NC emergence.

K Biological Details

Macaque IT (Cadieu et al., 2014) ($K = 7$): per-image $r = 0.41$; V4 $r = 0.116$ (hierarchy gradient). Mouse V1 (Stringer et al., 2019) ($K = 11$): per-image $r = 0.64$. Allen Neuropixels: $K = 118$, 32 mice, response window 50–250ms. Population geometry dominated by $\approx$10 effective dimensions (Pospisil and Pillow, 2025). Universal dimensions: Chen and Bonner (2025) show diverse networks converge on < 10 latent dimensions aligned with brain, consistent with $d_{\text{eff,comp}} \approx 2$.

L Scaling Dynamics

Power-law fit $\kappa \propto N^{\gamma}$: $\hat{\gamma} = 0.003$, $R^2 = 0.008$, $p = 0.91$ — κ_{nearest} does not scale with model size. MAP@10 does scale ($R^2 = 0.90$, $p = 0.05$).